AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2025

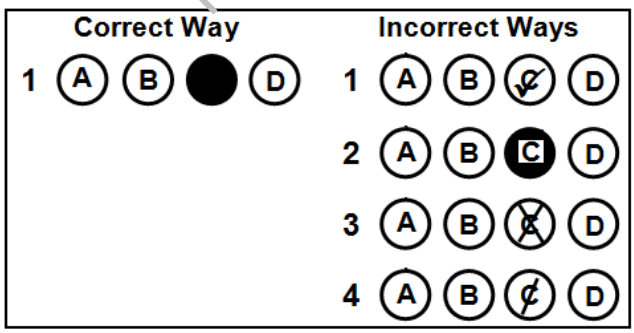
Physics Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. Three capacitors of different capacitances are connected in series.

   Which of the following quantities will remain the same across all the capacitors?

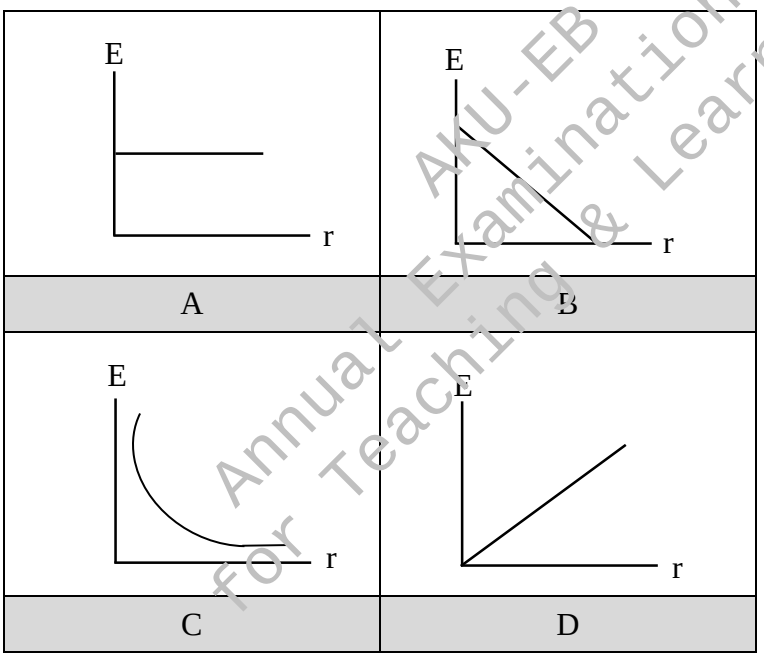
   A. Charge
   B. Voltage
   C. Reactance
   D. Capacitance

2. The working principle of an inkjet printer is based on



   A. electricity.
   B. electronics.
   C. electrostatics.
   D. electromagnetism.

3. Which of the following graphs correctly represents the relationship between the electric field (E) and the position (r) of a point charge?



4. In a photocopier, when light is projected onto a selenium coated drum, it

   A. emits light.
   B. changes colour.
   C. becomes electrically change.
   D. conduct electricity and loses its charge.

5. If two points are at the same potential in an electric field, then the work done in moving a unit positively charged from one point to another is equal to

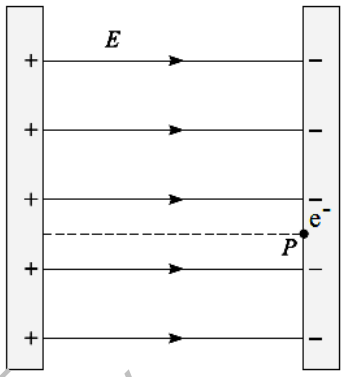
   A. – 0.5 J.
   B. 0 J.
   C. 1 J.
   D. 2 J.

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6. An electron is released from rest at point *P* on the negative plate as shown in the given diagram.

The direction of the force and the acceleration on the electron will be

	Force	Acceleration
A	leftward	leftward
B	leftward	rightward
C	rightward	leftward
D	rightward	rightward



7. How much power is dissipated by a resistor of resistance 10 Ω when current of 10 A is passing through it?

A. 1 W
B. 10 W
C. 100 W
D. 1,000 W

8. Which of the following BEST describe the I-V characteristics of a semi-conductor diode?

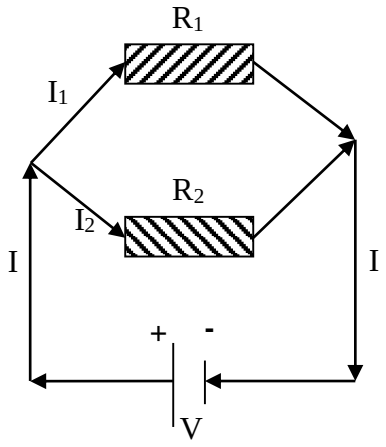
A. Current is directly proportional to the applied voltage.
B. Current is inversely proportional to the applied voltage.
C. Current exponentially increases with the applied voltage.
D. Current exponentially decreases with the applied voltage.

9. If the terminal potential difference of a battery is equal to its electromotive force (e.m.f.), then the internal resistance of the battery becomes

A. zero.
B. infinite.
C. constant.
D. maximum.

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10. The given circuit diagram is an example of



A. Lenz's law.
B. Faraday's law.
C. Kirchoff's law.
D. Coulomb's law.

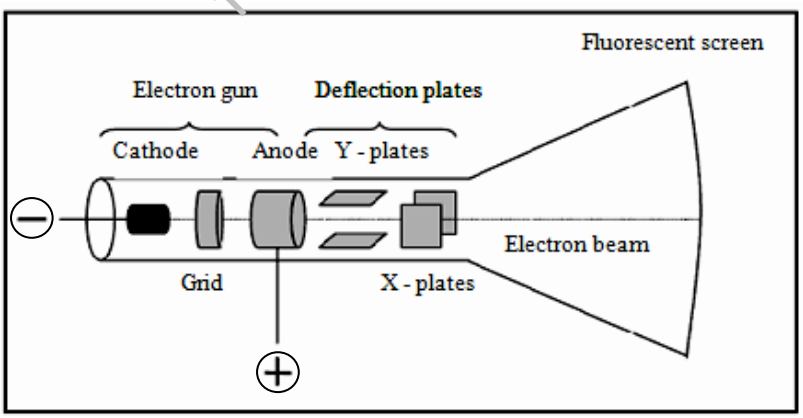
11. An electric device that is used to measure the value of an unknown resistance is a

A. rheostat.
B. ammetre.
C. voltmetre.
D. Wheatstone bridge.

12. If an electron is allowed to move perpendicular to the uniform magnetic field, then it will follow the direction in a

A. zig-zag path.
B. circular path.
C. straight path.
D. triangular path.

13. The function of the grid in the given cathode ray oscilloscope (CRO) is to



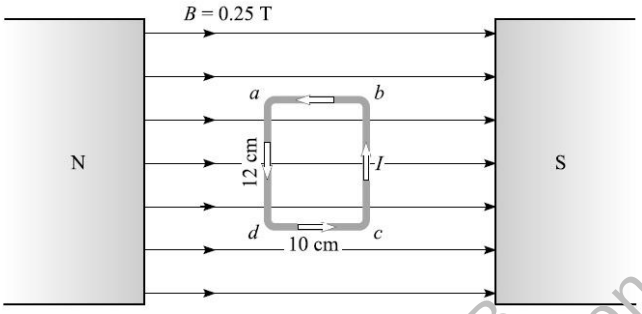
A. eject electrons.
B. deflect electrons.
C. accelerate electrons.
D. control the number of electrons.

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14. A current carrying coil is placed in a uniform magnetic field. If the torque produced in the coil becomes double, then the

    A. area will be doubled.
    B. current will be quadrupled.
    C. length will reduce to one half.
    D. moment arm will reduce to one half.

15. If a coil having 40 loops carries a current of 2.0 A in a magnetic field B = 0.25 T, then the force on each vertical wire is



    A. 0.03 N.
    B. 0.06 N.
    C. 1.20 N.
    D. 2.40 N.

16. The physical quantity that measures by voltmetre is

    A. current.
    B. resistance.
    C. conductance.
    D. potential difference.

17. The law that is used to determine the direction of electric current in an alternating current (AC) generator is

    A. Lenz's law.
    B. Faraday's law.
    C. Ampere's law.
    D. Coulomb's law.

18. Which of the following phenomena occurs when an electromotive force (e.m.f.) is induced in the coil of an alternating current (AC) generator?

    A. Self-induction
    B. Mutual-induction
    C. Electrostatic induction
    D. Electromagnetic induction

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19. From the following listed parts, the main components of a transformer are

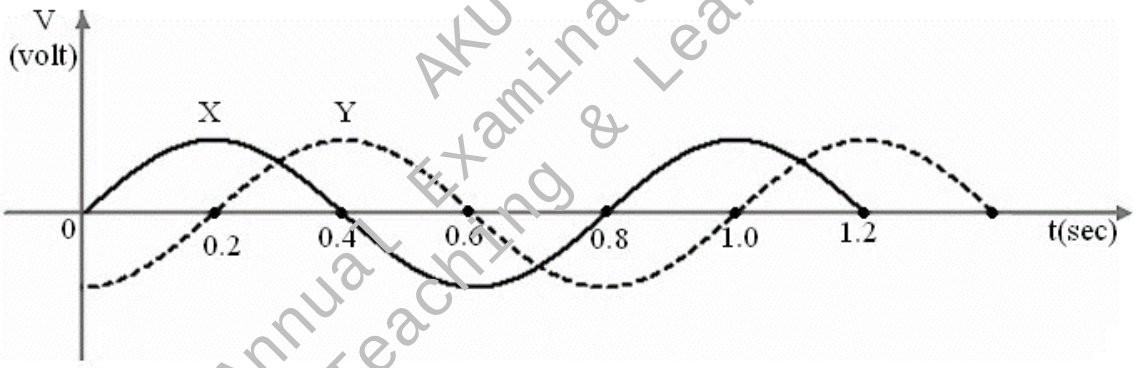
I. iron core
II. slip rings
III. primary coil
IV. carbon brushes

A. I and III.
B. I and IV.
C. II and III.
D. II and IV.

20. A transformer steps down from 220 V to 110 V. If the number of turns in the secondary coil is 40, then the number of turns in the primary coil is

A. 20
B. 80
C. 120
D. 160

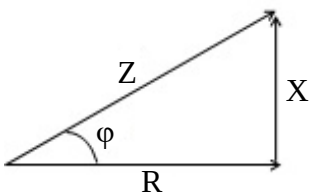
21. The given diagram shows two sinusoidal voltages X and Y of the same frequency.



The phase relationship between X and Y is

A. X lead Y by π/4.
B. Y lead X by π/4.
C. X lead Y by π/2.
D. Y lead X by π/2.

22. In the given diagram of an AC series circuit, if φ is the phase angle of impedance, then the value of φ will be calculated by



A. $\tan^{-1} X_c / R$.
B. $\tan^{-1} X_L / R$.
C. $\tan^{-1} R / X_c$.
D. $\tan^{-1} R / X_L$.

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23. If the peak voltage of an A.C. is 120 V, then its root mean square value (V_{rms}) is

A. 60.0 V.
B. 84.8 V.
C. 169.7 V.
D. 240.0 V.

24. The phase difference between each pair of coils of a three-phase alternating current (AC) generator is

A. 0°
B. 90°
C. 120°
D. 240°

25. Liquids with the frozen disordered structure are the example(s) of

I. polymeric solids
II. crystalline solids
III. amorphous solids

A. I only.
B. III only.
C. I and II.
D. II and III.

26. The ratio of tensile stress to tensile strain is called

A. Bulk modulus.
B. Shear modulus.
C. Young's modulus.
D. Modulus of rigidity.

27. Which of the following is created, when a thermal excitation occurs in a doped semi-conductor?

A. A hole
B. A positron
C. An electron
D. An electron-hole pair

28. The substance(s) in which the atoms form magnetic dipole is/ are

I. ferromagnetic
II. paramagnetic
III. diamagnetic

A. I only.
B. III only.
C. I and II.
D. II and III.

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29. The Curie point shows a relationship between

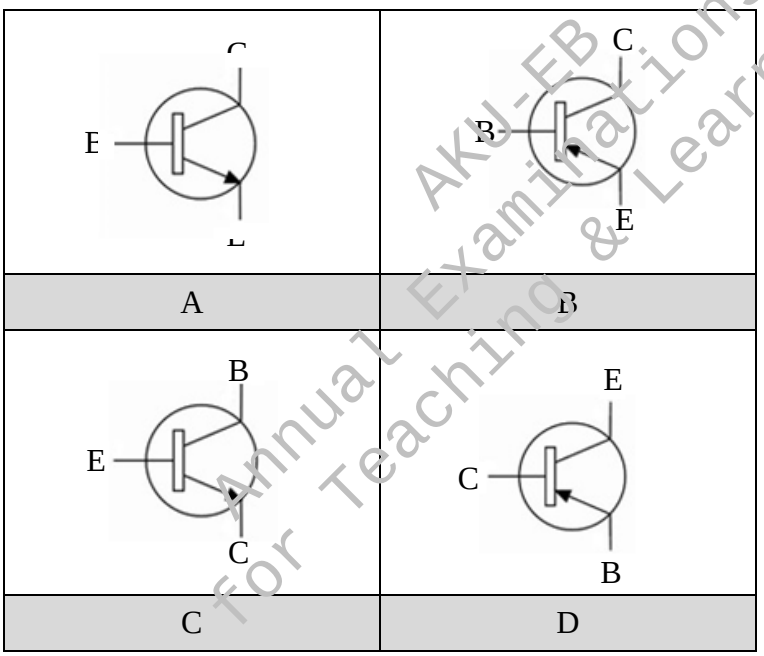
A. pressure and magnetism.
B. pressure and electric flux.
C. temperature and magnetism.
D. temperature and electric flux.

30. The device that exhibits the characteristic of allowing electric current to flow in only one direction is a

A. diode.
B. buffer.
C. transistor.
D. resistor-transistor.

31. The diagram depicting p-n-p transistor is

(**Note**: B, C and E represent base, collector and emitter respectively.)



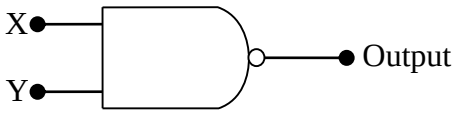
32. In a non-inverting amplifier the feedback resistor (R_2) will be connected between the

A. input and ground terminal.
B. output and inverting terminal.
C. output and non-inverting terminal.
D. inverting terminal and ground terminal.

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33. The truth table and the given symbol represents a/ an

X	Y	Output
0	0	1
0	1	1
1	0	1
1	1	0



A. OR gate.
B. NOR gate.
C. AND gate.
D. NAND gate.

34. The logic gate which inverts the input signal is a/ an

A. OR gate.
B. NOT gate.
C. NOR gate.
D. AND gate.

35. If the temperature of black body increases, then the wavelength corresponding to maximum intensity will

A. become zero.
B. remain the same.
C. shift to longer wavelength.
D. shift to shorter wavelength.

36. In case of photoelectric emission, stopping potential for a metal surface depends on the

A. intensity of incident light.
B. frequency of incident light.
C. work function for metal surface.
D. threshold frequency for the metal surface.

37. Albert Einstein's special theory of relativity shows that mass is a form of

A. force.
B. speed.
C. charge.
D. energy.

38. When a metal surface is exposed to light, the energy of the ejecting electrons does NOT depend on the

A. energy of incident photon.
B. intensity of incident photon.
C. frequency of incident photon.
D. wavelength of incident photon.

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39. In quantum mechanics, the uncertainty in the momentum and position of a particle is due to

A. its low velocity.
B. the duality of matter.
C. a certain wavelength.
D. its two-dimensional motion.

40. The hydrogen atom does NOT emit X-rays. The reason behind this is that the

A. energy levels in a hydrogen atom are too far apart.
B. hydrogen atom lacks enough protons to produce X-rays.
C. X-rays can only be emitted by atoms with multiple electrons.
D. energy levels in a hydrogen atom are too small apart to produce X-rays.

41. In Bohr's atomic model of the hydrogen atom, the radius of the third orbit is greater than that of the first orbit by

A. 3 times.
B. 6 times.
C. 9 times.
D. 12 times.

42. When a hydrogen atom radiates energy spectral lines are formed. This is because the electrons of hydrogen atom

A. change their orbital motion.
B. have weak nuclear attractive force.
C. shift from ground state to excited state.
D. jump from higher to a lower energy level.

43. The total energy of an electron in an orbit, around the nucleus, is the sum of

A. potential and kinetic energy.
B. rotational energy and kinetic energy.
C. excitation energy and potential energy.
D. rotational energy and excitation energy.

44. All of the following are properties of X-rays EXCEPT that they

A. are diffracted by a crystal lattice.
B. pass through wood and flesh easily.
C. are deflected by electric and magnetic fields
D. travel in a straight line with the velocity of light.

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45. The basic particle of an atom that CORRECTLY corresponds the mass and charge of a positron is

	Mass of Positron	Charge of Positron
A	Proton	Electron
B	Electron	Proton
C	Neutron	Proton
D	Neutron	Electron

46. In the given chemical equation, X represents

$^{2}H_{1} + {}^{2}H_{1}$ ---------> $^{3}He_{2} + X$

A. an alpha particle.
B. a beta particle.
C. a neutron.
D. a proton.

47. The nuclear radiation(s) possess(es) the maximum velocity is/ are

I. alpha particles
II. beta particles
III. gamma rays

A. II only.
B. III only.
C. I and II.
D. I and III.

48. The detector(s) that can operate(s) at high voltage is

I. Geiger counter
II. Solid state detector
III. Wilson cloud chamber

A. I only.
B. II only.
C. I and III.
D. II and III.

49. The force that is responsible for confining neutrons and protons within the nucleus is a

A. gravitational force.
B. weak nuclear force.
C. strong nuclear force.
D. electromagnetic force.

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50. The sub-atomic particle that experiences all the four fundamental forces is

A. quark.
B. boson.
C. lepton.
D. fermion.

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